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CMSE 801 - Day 5

Getting more comfortable with writing functions
through practice

Jan 27, 2026



General Course Announcements

- Reminder: Homework 1 is available on the course website. This assignment will be **due at 11:59pm on Saturday, Feb 21.**
 - This is an opportunity to showcase all of the Python fundamentals you've picked up in course and to practice finding useful examples and resources on the web.
 - **The assignment might end up feeling a little tedious (sorry!) but it should be useful and nothing too challenging should come up.** *Give yourself time to work through and consider doing it in multiple sittings.*
 - There is a D2L submission folder for when you complete the assignment

General Course Announcements (cont'd)

- AI Course Assistant is now online! You should have all received an email invitation from **HiTA**.
 - HiTA employs a technology called **Retrieval-Augmented Generation (RAG)** to provide context to a **Large Language Model (LLM)** based on course materials, instructor's instructions, and conversation history
 - Let me know if you have trouble logging in!
- Reminder that office hours and locations are on the website:
<https://cmse.msu.edu/cmse801>
- Remember: at the end of class **you will be uploading your in-class notebook to D2L.**

Struggles from Day 5 PCA

- Mostly the last question (Part 5) -- **we'll look at that at the end**
- setting default values in functions
 - still some confusion about variable scope -- see Day 4 slides for more details
- calling the function

Let's look at some of the things we can do with functions by taking the simple "Happy Birthday" example

In [1]:

```
# Define a function that prints "Happy Birthday to <person>" where the person is a name argument I pass i  
def happy_birthday(name):  
    print("Happy birthday to", name)
```

In [2]:

```
# here we are calling the function we just defined  
happy_birthday("Ian")
```

Happy birthday to Ian

In [3]:

```
# What if I want there to be a default name?  
def happy_birthday(name='you!'):  
    print("Happy birthday to", name)
```

In [4]:

```
# calling this function again  
happy_birthday()
```

Happy birthday to you!

In [5]:

```
# What if I want to return the statement instead of just print it?  
def happy_birthday(name='you!'):   
    return("Happy birthday to"+" "+name)
```

In [7]:

```
wishes = happy_birthday("Sparty")  
print(wishes)
```

Happy birthday to Sparty

Let's work through Part 5...

Do the following:

1. Write function called `x_squared`.
2. The function should take in a list of x -values of any length, but should have a default of `[1,2,3]`.
3. The function should **return** a new list of values that are x^2 values of the original input values.
4. Print the resulting new list.

In [8]:

```
def x_squared(x_vals=[1,2,3]):  
    x2 = []  
    for x in x_vals:  
        x2.append(x**2)  
    return x2
```


In [9]:

```
# Let's test it out  
xs = [2,4,6,8,10]  
  
squared = x_squared(xs)  
print(squared)
```

[4, 16, 36, 64, 100]

In [10]:

```
# Let's also confirm that my default ([1,2,3]) is working:  
default_squared = x_squared()  
print(default_squared)
```

[1, 4, 9]

What if we wanted to use a different function *inside* this function?

For example, I could use my `happy_birthday` function from before (though it might not be the obvious choice...)

In [11]:

```
def x_squared(vals=[1,2,3]):  
    x2 = [] # Initialize the list to return  
    for x in vals: # Looping by ???  
        x2.append(x**2)  
        if x**2 > 50:  
            print(happy_birthday("someone!"))  
    return x2
```

In [12]:

```
squared = x_squared([2,4,6,8,10])  
print(squared)
```

```
Happy birthday to someone!  
Happy birthday to someone!  
[4, 16, 36, 64, 100]
```

In [13]:

```
# I could also pass a default value for when I call the happy_birthday function
def x_squared(vals=[1,2,3], hb='someone else!'):
    x2 = [] # Initialize the list to return
    for x in vals: # Looping by ???
        x2.append(x**2)
        if x**2 > 50:
            print(happy_birthday(hb))
    return x2
```

In [14]:

```
squared = x_squared([2,4,6,8,10])
print(squared)
```

```
Happy birthday to someone else!
Happy birthday to someone else!
[4, 16, 36, 64, 100]
```

Goals for today

- Practice writing functions to structure and compartmentalize your code and make it **reusable**.
 - This will sometimes involve using functions *inside* other functions
- Experiment with how you can use function to start building a "rideshare app" (not really, but go with it..)

Looking forward - Day 6 PCA: Using modules in Python (specifically numpy and matplotlib)

- Explore how you can "import" modules into your notebooks to gain access to additional packages and new functionality
 - `matplotlib` -- finally we can start visualizing values and data! this is the most common library for making plots in Python
 - `numpy` -- you'll primarily start using it to make it easier to do math in Python, but we'll also use it to make working with data easier